Preliminary Research Question and Dataset

1. Research question

2. Data collection

Our project combines data from several sources in order to build a regional dataset for Ukraine.

Education data

For the education part, we use data from the State Statistics Service of Ukraine.

Instead of downloading each indicator separately, we downloaded the complete dataset on educational institutions. The raw dataset contains data for different education levels and several types of breakdowns.

The education data cover the period 2019–2025 and initially contain:

- 26,575 rows
- 23 columns
- 38 education indicators

The indicators cover pre-school, general secondary, vocational, professional pretertiary, tertiary and postgraduate education.

Conflict data

Socioeconomic data

The three data sources will later be combined using common region and year variables.

3. Quantitative description of the education dataset

We first checked the temporal and regional coverage of the 38 education indicators.

Based on their relevance and data availability, we selected 19 indicators covering five education levels:

Education level	Main variables retained
Pre-school	children, institutions, places, employees
General secondary	pupils, institutions, teachers
Vocational and technical	pupils/listeners, institutions, enrolled students, graduates
Professional pretertiary	students, institutions, enrolments, graduates
Tertiary	students, institutions, enrolments, graduates

After preprocessing, the regional education dataset contains:

- 25 regions
- 7 years (2019–2025)
- 19 indicators

- 3,325 region-year-indicator observations

Among these observations:

- 3,124 values are available
- 201 values are missing
- the missing-value rate is approximately 6.05%

For the future merge with the other project data, we also created a wide-format dataset containing:

- 175 region-year observations
- 19 education variables

This corresponds to 25 regions observed over 7 years.

4. Variables and their meaning

The selected education variables describe different parts of the education system.

Students / pupils: measure the number of people studying at each education level.

Institutions: measure the number of educational establishments.

Enrolments: represent the number of people entering an education level during a given period.

Graduates: represent the number of people completing an education level.

For pre-school education, we also keep the **number of available places** and the **number of full-time employees.**

These variables were selected because they allow us to study both participation in education and changes in the education infrastructure.

5. Preprocessing and choices

The raw education data require several preprocessing steps.

First, the indicators do not all have the same structure. Some are divided by terrain type, educational degree, staff position or other categories, while others already contain a direct total value.

Because of this, we could not apply exactly the same filter to every indicator. For each selected variable, we kept the most general available value in order to obtain one observation per region and indicator.

We then transformed the data from the original format, where years were stored in separate columns, into a region-year-indicator-value format. Shorter and more consistent variable names were also created.

The national observation “Ukraine” was kept separately because it represents a country total rather than a region.

Crimea and Sevastopol were excluded from the regional dataset because the main education indicators are systematically missing for these areas.

Other missing observations were kept as missing and were not replaced by zero, since a missing value does not mean that the real value is zero.

Some indicators were also excluded from the core dataset. For example, indicators concerning foreign students and postgraduate education have less consistent regional coverage and are less directly related to our main research question.

Professional pretertiary and tertiary teacher indicators were not kept in the current core dataset because their staff breakdowns do not provide a clear single overall total.